

Discrete Probability Distributions

Method Dossier with Field-Specific Examples | Basic Level

1. Method Description

This dossier covers four key discrete distributions: Binomial $B(n,p)$, Poisson $Po(\lambda)$, Hypergeometric $H(N,K,n)$, and Geometric $Geo(p)$.

Binomial: Fixed trials n , constant success prob p . $P(X=k) = C(n,k) \cdot p^k \cdot (1-p)^{(n-k)}$.

Poisson: Rare events in fixed interval. $P(X=k) = \frac{\lambda^k}{k!} \cdot e^{-\lambda}$.

Hypergeometric: Sampling without replacement from finite population.

Geometric: Number of trials to first success.

2. Field-Specific Examples

Social Sciences

Binomial: $n=1,000$ survey, 62% satisfied. Probability of 600+ if true rate = 58%?

Economics

Poisson: Store averages 5 customers per 10 min. Probability of 8+ in next 10 min?

Medicine

Hypergeometric: Lot of 50 vaccine doses, 5 contaminated. Sample of 10: $P(\text{exactly 1 contaminated})$?

Sports Sciences

Geometric: Free throw success rate 75%. Expected trials until first hit = $1/0.75 = 1.33$.

3. Assumptions

- Binomial: fixed n , independence, constant p , 2 outcomes
- Poisson: rare, independent events, constant rate λ
- Hypergeometric: sampling without replacement, finite population
- Geometric: independent trials, constant p , stop at first success

4. Code Examples

R:

```
dbinom(3,10,0.3)
ppois(2,3)
dhyper(1,5,45,10)
```

Python:

```
from scipy.stats import binom, poisson, hypergeom
binom.pmf(3,10,0.3)
poisson.cdf(2,3)
```

5. Extension

Normal approximation to Binomial

If $np > 5$ and $n(1-p) > 5$: $B(n,p)$ approx $N(np, np(1-p))$.

Negative Binomial

Number of failures before r -th success.

Multinomial

Extension of binomial to 3+ categories.

6. Exercises

Exercise 1: MC test 20 questions, 4 options each. X = correct by guessing. a) Distribution type? b) $P(X \geq 8)$? c) $P(X = 10)$?

Exercise 2: ER averages 2.4 emergencies per hour. Poisson. a) $P(X=0)$ in 1 hour? b) $P(X>4)$ in 2 hours?

Exercise 3: Shipment of 20 devices, 3 defective. Sample 5 w/o replacement. $P(\text{exactly 1 defective})$?